

YEAR 7 MATHEMATICS · CHAPTER 1

Integers

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|--------------------------------------|--|
| 1.01 Integers and opposites | 1.02 The number line |
| 1.03 Comparing and ordering integers | 1.04 Adding integers |
| 1.05 Subtracting integers | 1.06 Multiplying integers |
| 1.07 Dividing integers | 1.08 Order of operations with integers |
| 1.09 Integers on a calculator | 1.10 Integer problems |

Name

Class

Teacher

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a graph, dot plot or stem-and-leaf plot. Use the grid or template given and a ruler.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

NSW Mathematics K-10 Syllabus (2022), Stage 4 Number and algebra: MA4-INT-C-01 compares, orders and calculates with integers to solve problems.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
1.01

Integers and opposites

Date _____

Learning intentions

We are learning to:

- understand what an integer is
- write the opposite of an integer

Success criteria

I can:

- ☐ describe a real situation with a positive or negative number
- ☐ write the **opposite** of any integer
- ☐ explain why 0 is its own opposite

Key terms Write the definition as your teacher explains it.

Integer

Negative number

Opposite

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Write an integer for: a temperature 6°C below zero; a gain of 4 metres.**Example 2** Write the opposite of: -9 , 15 , 0 **YOU DO** On your own: Set A. Write an integer for each situation.1 5°C below zero

4 a rise of 8 floors

2 a profit of \$40

5 a loss of 3 kg

3 12 m below sea level

6 200 m above sea level

YOU DO On your own: Set B. Write the opposite of each integer.

1 -2

4 -10

2 -23

5 8

3 56

6 0

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** A lift goes 3 floors down, then 3 floors up. Write both moves as integers. Where does it end?**Example 4** Which is further from 0: -8 or 6? Explain using opposites.**YOU DO** On your own: Set C. Opposites in real life.

1 What is the opposite of “3 steps forward”? Write both as integers.

4 Which is further from 0: -12 or 9?

2 A diver is at -14 m. What is the opposite position?

5 Write two integers that are 5 units from 0.

3 Write the opposite of the opposite of -7.

6 A bank balance of -\$60 means what?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** The opposite of a number is 25. What is the number?**Example 6** List the integers whose distance from 0 is exactly 7.
YOU DO On your own: Set D. Think it through.**1** Is -0 different from 0? Explain.
3 List all integers whose distance from 0 is less than 3.
2 The opposite of a number is 18. What is the number?
4 Does every integer have an opposite? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write the opposite of -15 .
Level 2Write an integer for "a drop of 9°C ", then write its opposite.
Level 3The opposite of a number is -31 . What is the number? Explain.

 Extension · Stretch yourself

 Worked example

The opposite of n is written $-n$. If $n = -4$, what is $-n$?

- $-n$ means “the opposite of n ”
- The opposite of -4 is 4

Answer: $-n = 4$

YOU DO

On your own: read the worked example, then try these.

- E1** If $n = -11$, find $-n$ and $-(-n)$.

- E2** Is $-n$ always a negative number? Give an example.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Integers** are whole numbers: ..., -2 , -1 , 0 , 1 , 2 , ...
 2. Negative numbers are below zero.
 3. The **opposite** is the same distance from 0 on the other side.
 4. 0 is its own opposite.

 Level 1

Write -7 as a real situation.

1. Negative means below or less

Answer: 7°C below zero

 Level 2

Opposite of -23 ?

1. Same distance, other side of 0

Answer: 23

 Level 3

Which integers are 4 from 0?

1. One on each side of 0

Answer: 4 and -4

 Extension

If $n = -6$, find $-n$.

1. Opposite of -6

Answer: 6

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LESSON
1.02

The number line

Date _____

Learning intentions

We are learning to:

- show integers on a number line
- find integers between two numbers

Success criteria

I can:

- ☐ draw a number line with an even scale
- ☐ mark positive and negative integers in the right places
- ☐ list the integers **between** two numbers

Key terms Write the definition as your teacher explains it.

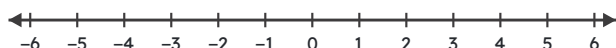
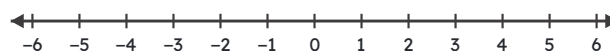
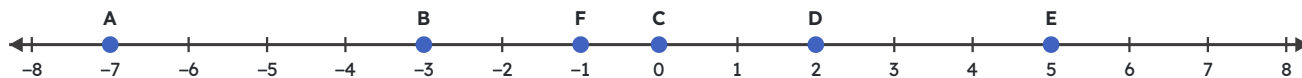
Number line

Positive

Negative

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Mark 3, -2, 0 and -5 on the number line.**Example 2** Mark the integers between -3 and 2.**YOU DO** On your own: Set A. Which integer is shown by each letter?

1 A

4 D

2 B

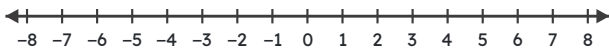
5 E

3 C

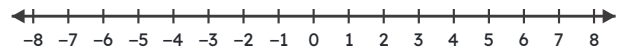
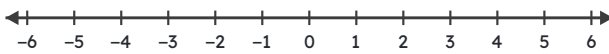
6 F

YOU DO On your own: Set B. Mark each set of numbers on the number line.

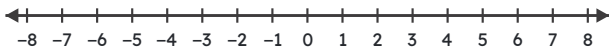
1 6, -3, 5, 1, -1



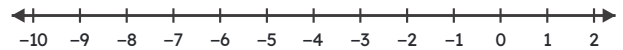
2 -7, 4, -4, 0

**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** Mark -4, 2, -1 and 5.**Example 4** Which integer is 3 to the left of -2?**YOU DO** On your own: Set C. Mark each set on the number line.

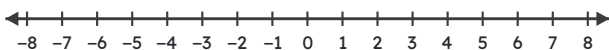
1 integers between -2 and 5



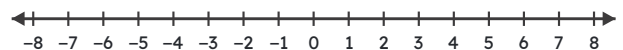
3 integers between -8 and -3



2 integers less than 1 and greater than -6



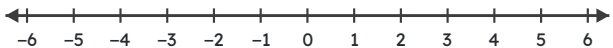
4 all even integers from -6 to 6





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Mark -30 , 10 and -15 using steps of 5 .**Example 6** How many integers are between -6 and 4 ?**YOU DO** On your own: Set D. Number lines with a scale.**1** Mark -20 , 15 , -5 and 30 . Use steps of 10 .**3** How many integers are between -9 and 9 ?**2** Mark the integers more than -4 but at most 2 .**4** Which integer is halfway between -10 and 4 ?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1List the integers between -3 and 1 .
Level 2Which is further left on a number line: -8 or -2 ?
Level 3Which integer is halfway between -9 and 3 ?

Extension · Stretch yourself

Worked example

How many integers are between -15 and 12 ?

1. From -14 to -1 : 14 integers
2. Then 0: 1 integer
3. From 1 to 11: 11 integers

Answer: $14 + 1 + 11 = 26$ integers

YOU DO

On your own: read the worked example, then try these.

- E1** How many integers are between -30 and 25 ?

- E2** Which integer is halfway between -17 and 5 ?

Lesson summary

Missed the lesson? Start here.

Remember

1. Numbers **increase** to the right.
2. Negatives are left of 0.
3. Use an **even scale**: equal gaps.
4. "Between" does not include the end numbers.

Level 1

Where is -3 ?

1. 3 steps left of 0

Answer: 3 marks left of zero

Level 2

Integers between -2 and 2 ?

1. Do not include -2 and 2

Answer: $-1, 0, 1$

Level 3

Mark -20 on a scale of 10s

1. Each gap is 10
2. 2 gaps left of 0

Answer: second mark left of 0

Extension

Halfway between -6 and 2 ?

1. Distance 8, half is 4
2. $-6 + 4$

Answer: -2

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LESSON
1.03

Comparing and ordering integers

Date _____

Learning intentions

We are learning to:

- compare integers using $<$ and $>$
- put integers in ascending and descending order

Success criteria

I can:

- ☐ use a number line to decide which integer is larger
- ☐ write true statements with $<$ and $>$
- ☐ order a set of integers from smallest to largest and largest to smallest

Key terms Write the definition as your teacher explains it.

Ascending

Descending

 $<$ and $>$

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

True or false: $-4 > 2$? $-2 < -3$?

Example 2

Write in ascending order: 3, -6, 0, -1, 5

YOU DO

On your own: Set A. Write true (T) or false (F).

1 $-4 > 2$

2 $-2 < -3$

3 $2 > -1$

4 $-11 < -10$

5 $-3 > -4$

6 $-9 > -2$

YOU DO On your own: Set B. Write $<$ or $>$ between each pair.

1 $5 \square -5$

4 $-12 \square -21$

2 $-7 \square -1$

5 $-1 \square 1$

3 $0 \square -3$

6 $-100 \square -99$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Write $<$ or $>$: $-6 \square -9$ **Example 4** Descending: $-2, 8, -15, 0, 3$ **YOU DO** On your own: Set C. Order each set.1 Ascending: $4, -2, 7, -8, 0$ 4 Descending: $-7, -2, -11, 6, -1$ 2 Descending: $-9, 14, 3, 0, -5, -10, -1$ 5 Ascending: temperatures $-4^{\circ}\text{C}, 3^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}, -1^{\circ}\text{C}$ 3 Ascending: $-15, 12, -3, -20, 8$ 6 Descending: heights $40\text{ m}, -8\text{ m}, -35\text{ m}, 12\text{ m}$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Ascending: $-(-3)$, -5 , 1 , $-(-2)$

Example 6 Write 3 integers between -12 and -8 in ascending order.
YOU DO On your own: Set D. Challenge.

- 1 Write 3 integers that are less than -5 but greater than -10 .

- 3 Put in ascending order: -1 , -10 , -100 , -1000 , 0

- 2 Which is the smallest: $-(-6)$, -8 or $-(-10)$?

- 4 Is it true that the larger the number after a negative sign, the smaller the integer? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1True or false: $-7 > -3$?
Level 2Write in descending order: 2 , -5 , 0 , -1

Level 3Write in ascending order: -12 , 5 , $-(-8)$, -20 , 0

Extension · Stretch yourself

Worked example

Write in ascending order: $-3, 2, -(-5), -7, 0$

- $-(-5)$ is the opposite of -5 , which is 5
- Now order: $-7, -3, 0, 2, 5$

Answer: $-7, -3, 0, 2, -(-5)$

YOU DO

On your own: read the worked example, then try these.

- E1 Order from smallest: $-(-2), -4, -(-4), 3, -(-9)$

- E2 Find two integers between $-(-3)$ and $-(4)$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Further **right** on the number line = **larger**.
 2. $<$ means “is less than”; $>$ means “is greater than”.
 3. **Ascending**: smallest to largest.
 4. **Descending**: largest to smallest.

Level 1

Is $-5 < -1$?

1. -5 is further left

Answer: True

Level 2

Order $-2, 4, -8, 1$

1. Smallest is the most negative

Answer: $-8, -2, 1, 4$

Level 3

Descending: $-3, -30, 3, 0$

1. Largest first

Answer: $3, 0, -3, -30$

Extension

Order $-(-4), -6, 2$

1. $-(-4) = 4$

Answer: $-6, 2, 4$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.04

Adding integers

Date _____

Learning intentions

We are learning to:

- add positive and negative integers

Success criteria

I can:

- ☐ use a number line to add integers
- ☐ know that adding a negative is the same as subtracting
- ☐ add three or more integers

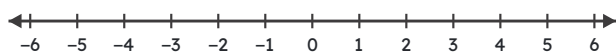
Key terms Write the definition as your teacher explains it.

Sum

Adding a negative

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $3 + (-4)$. Use the number line.**Example 2** $-8 + (-2)$ and $-4 + 8$ **YOU DO** On your own: Set A. Evaluate each sum.

1 $5 + (-2)$

2 $-3 + 7$

3 $-6 + (-1)$

4 $10 + (-13)$

5 $-5 + (-5)$

6 $-3 + 6$

YOU DO On your own: Set B. Evaluate each sum.

1 $-9 + 4$

4 $-1 + 15$

2 $8 + (-8)$

5 $20 + (-25)$

3 $-12 + (-3)$

6 $-7 + (-7)$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $-5 + (-4) + (-2)$

Example 4 $-8 + 5 + 4$

YOU DO On your own: Set C. Three or more integers.

1 $-5 + (-4) + (-2)$

4 $3 + (-9) + 10$

2 $-8 + 5 + 4$

5 $-20 + 15 + (-6)$

3 $-7 + 6 + (-7)$

6 $-1 + (-2) + (-3) + (-4)$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** It is -7°C and the temperature rises 12°C . New temperature?
Example 6 Find the missing number: $\square + 9 = -4$

YOU DO On your own: Set D. Adding integers in context.

- 1 The temperature is -6°C and rises 11°C . What is the new temperature?

- 3 Maya has \$25 and spends \$40 on credit. What is her balance?

- 2 A lift at level -2 goes up 7 floors. Which level is it at?

- 4 Find the missing number: $-9 + \square = -3$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-6 + 9$

Level 2

$-4 + (-8) + 5$

Level 3It is -3°C and the temperature drops 5°C , then rises 9°C . What is the temperature now?

Extension · Stretch yourself

Worked example

Find the missing number: $\square + (-14) = -5$

1. Think: what plus -14 gives -5 ?
2. Undo by adding 14: $-5 + 14 = 9$
3. Check: $9 + (-14) = -5$ ✓

Answer: 9

YOU DO

On your own: read the worked example, then try these.

E1 $\square + (-8) = -20$

E2 $-13 + \square + 6 = 0$

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding a **positive**: move right.
2. Adding a **negative**: move left.
3. Same signs: add and keep the sign.
4. Different signs: subtract, keep the sign of the bigger number.

Level 1

$4 + (-6)$

1. Start at 4, move 6 left

Answer: -2

Level 2

$-5 + (-3)$

1. Same signs: $5 + 3 = 8$, keep -

Answer: -8

Level 3

$-9 + 4 + (-2)$

1. $-9 + 4 = -5$
2. $-5 + (-2) = -7$

Answer: -7

Extension

$\square + (-4) = 1$

1. $1 + 4 = 5$

Answer: 5

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.05

Subtracting integers

Date _____

Learning intentions

We are learning to:

- subtract positive and negative integers

Success criteria

I can:

- ☐ subtract a positive integer by moving left
- ☐ know that subtracting a negative is the same as adding
- ☐ simplify expressions with + and – signs

Key terms Write the definition as your teacher explains it.**Difference****Subtracting a negative**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $7 - 11$ and $-4 - 8$ **Example 2** $3 - (-4)$ and $-5 - (-9)$ **YOU DO** On your own: Set A. Evaluate each expression.

1 $7 - 11$

4 $-6 - 1$

2 $-4 - 8$

5 $0 - 5$

3 $2 - 9$

6 $-10 - 10$

YOU DO On your own: Set B. Subtracting a negative: change to adding.

1 $3 - (-4)$

4 $8 - (-2)$

2 $-5 - (-9)$

5 $-12 - (-3)$

3 $-5 - (-5)$

6 $0 - (-7)$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $-3 - 4 - 5$ **Example 4** $8 - 6 - (-4)$ **YOU DO** On your own: Set C. Evaluate each expression.

1 $-3 - 4 - 5$

4 $3 - 8 + (-6)$

2 $-8 + 5 - 4$

5 $-2 - (-7) - 10$

3 $8 - 6 - (-4)$

6 $15 - (-5) + (-20)$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** The temperature rose from -9°C to 6°C .
By how much?
Example 6 Find the missing number: $-2 - \square = 6$

YOU DO On your own: Set D. Subtraction in context.**1** The minimum temperature was -5°C and the maximum was 11°C . By how much did it rise?
3 A submarine at -120 m rises 45 m . How deep is it now?
2 Find the difference between -18 m and 7 m .
4 What is 4 less than -9 ?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-3 - 6$

Level 2

$4 - (-9)$

Level 3The temperature went from -7°C to 12°C . How much did it rise?

 Extension · Stretch yourself

 Worked example

Find the missing number: $6 - \square = 10$

1. Subtracting a negative adds
2. $6 + 4 = 10$, so we subtracted -4
3. Check: $6 - (-4) = 10$ ✓

Answer: -4

YOU DO

On your own: read the worked example, then try these.

E1 $-3 - \square = 5$

E2 $\square - (-8) = -2$

Lesson summary

Missed the lesson? Start here.

Remember

1. Subtracting a **positive**: move left.
2. Subtracting a **negative** = adding the positive.
3. $- (-)$ becomes $+$.
4. Difference between two temperatures: higher – lower.

 Level 1

$5 - 9$

1. Start at 5, move 9 left

Answer: -4

 Level 2

$-2 - (-6)$

1. $= -2 + 6$

Answer: 4

 Level 3

$-4 - 3 - (-8)$

1. $-4 - 3 = -7$
2. $-7 + 8$

Answer: 1

 Extension

Rise from -8°C to 5°C

1. $5 - (-8) = 13$

Answer: 13°C

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
1.06

Multiplying integers

Date _____

Learning intentions

We are learning to:

- multiply positive and negative integers

Success criteria

I can:

- ☐ know the sign rules for multiplying
- ☐ multiply two integers
- ☐ multiply three integers by working in pairs

Key terms Write the definition as your teacher explains it.**Product****Sign rule**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

 $6 \times (-3)$ and -2×5

Example 2

 $-8 \times (-6)$ and $-6 \times (-4) \times (-1)$

YOU DO

On your own: Set A. Evaluate each product.

1 $6 \times (-3)$

2 -2×5

3 -11×4

4 $7 \times (-9)$

5 -3×8

6 $10 \times (-10)$

YOU DO On your own: Set B. Negative $\times$ negative.

1 $-8 \times (-6)$

4 $-12 \times (-2)$

2 $-5 \times (-5)$

5 $-1 \times (-20)$

3 $-7 \times (-3)$

6 $-9 \times (-4)$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $-2 \times 3 \times 7$ **Example 4** $-4 \times 2 \times (-5)$ **YOU DO** On your own: Set C. Evaluate each product.

1 $-6 \times (-4) \times (-1)$

4 $(-3)^2$

2 $-2 \times 3 \times 7$

5 $-5 \times (-2) \times 10$

3 $-4 \times 2 \times (-5)$

6 $(-2) \times (-2) \times (-2)$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find the missing number: $\square \times (-7) = 56$

Example 6 Is $(-1)^{100}$ positive or negative? Explain.
YOU DO On your own: Set D. Think it through.

- 1 A diver descends 4 m every minute. Where is she after 7 minutes?

- 3 Is the product of 5 negative numbers positive or negative? Explain.

- 2 Find the missing number: $-6 \times \square = 42$

- 4 The temperature drops 3°C each hour for 6 hours. What is the total change?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

-7×6

Level 2

$-8 \times (-9)$

Level 3

$-3 \times (-4) \times (-5)$

Extension · Stretch yourself

Worked example

Evaluate $(-1) \times (-1) \times (-1) \times \dots$ with 15 factors of -1 .

- Each pair of -1 s makes $+1$
- 15 factors = 7 pairs and one left over

Answer: -1

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $(-2)^5$

E2 Find two integers with a product of -24 and a sum of 5 .

Lesson summary

Missed the lesson? Start here.

Remember

- $++ = +$
- $+- = -$

2. $-x - = +$

4. Odd number of negatives $\rightarrow$ negative answer.

Level 1

$$4 \times (-7)$$

- Different signs $\rightarrow$ negative

Answer: -28

Level 2

$$-9 \times (-3)$$

- Same signs $\rightarrow$ positive

Answer: 27

Level 3

$$-2 \times (-5) \times (-3)$$

- $-2 \times (-5) = 10$
- $10 \times (-3)$

Answer: -30

Extension

$$(-3)^3$$

- Three negatives $\rightarrow$ negative
- 27

Answer: -27

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LESSON
1.07

Dividing integers

Date _____

Learning intentions

We are learning to:

- divide positive and negative integers

Success criteria

I can:

- ☐ use the same sign rules for dividing as for multiplying
- ☐ divide two integers
- ☐ work left to right with two divisions

Key terms Write the definition as your teacher explains it.**Quotient****Sign rule**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $-12 \div (-4)$ and $15 \div (-5)$ **Example 2** $24 \div (-6) \div (-2)$ **YOU DO** On your own: Set A. Evaluate each quotient.

1 $-12 \div (-4)$

4 $16 \div (-4)$

2 $15 \div (-5)$

5 $-24 \div 12$

3 $-6 \div (-6)$

6 $-1000 \div 10$

YOU DO On your own: Set B. Evaluate each quotient.

1 $-45 \div 9$

4 $-100 \div (-25)$

2 $-56 \div (-8)$

5 $0 \div (-3)$

3 $63 \div (-7)$

6 $-81 \div 9$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $-36 \div 3 \div 4$ **Example 4** $-64 \div (-4) \div 2$ **YOU DO** On your own: Set C. Work from left to right.

1 $24 \div (-6) \div (-2)$

4 $-60 \div (-5) \div (-3)$

2 $-36 \div 3 \div 4$

5 $100 \div (-10) \div (-5)$

3 $-64 \div (-4) \div 2$

6 $-48 \div 2 \div (-3)$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** A debt of \$96 is shared by 8 people.
Write each share as an integer.
Example 6 Find the missing number: $\square \div (-6) = -9$

YOU DO On your own: Set D. Division in context.**1** A \$-84 debt is shared equally by 4 people. What is each share?
3 Find the missing number: $\square \div (-5) = 7$

2 The temperature fell 18°C over 6 hours. What was the change each hour?
4 Find the missing number: $-72 \div \square = 9$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-36 \div 9$

Level 2

$-42 \div (-6)$

Level 3

$-80 \div (-4) \div (-5)$

 Extension · Stretch yourself

 Worked example
Evaluate $-6 \times 8 \div (-12)$

1. Work left to right
2. $-6 \times 8 = -48$
3. $-48 \div (-12) = 4$

Answer: 4

YOU DO

On your own: read the worked example, then try these.

E1 $-90 \div (-3) \times (-2)$

E2 $-5 \times (-12) \div (-4) \div 3$

Lesson summary

Missed the lesson? Start here.

Remember

1. Same signs $\rightarrow$ positive.
2. Different signs $\rightarrow$ negative.
3. 0 divided by any number is 0.
4. Two divisions: work left to right.

 Level 1

$-20 \div 4$

1. Different signs

Answer: -5

 Level 2

$-35 \div (-7)$

1. Same signs

Answer: 5

 Level 3

$-40 \div (-2) \div (-5)$

1. $-40 \div (-2) = 20$
2. $20 \div (-5)$

Answer: -4

 Extension

$\square \div (-3) = -6$

1. $-6 \times (-3)$

Answer: 18

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LESSON
1.08

Order of operations with integers

Date _____

Learning intentions

We are learning to:

- use the order of operations with positive and negative numbers

Success criteria

I can:

- ☐ do brackets first, then $\times$ and $\div$, then $+$ and $-$
- ☐ work left to right for operations of the same kind
- ☐ evaluate an expression with a fraction bar

Key terms Write the definition as your teacher explains it.**Order of operations****Fraction bar**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $-6 + (-3) \times 2$ **Example 2** $(9 - 17) \div (-4 + 6)$, written as a fraction**YOU DO** On your own: Set A. Evaluate each expression.

1 $4 - 3 \times 6$

4 $-2 + 5 \times 3$

2 $-12 \div 4 - 10$

5 $10 - (-2) \times 4$

3 $30 \div (-6) + 5$

6 $-8 \div 2 + 1$

YOU DO On your own: Set B. Evaluate each expression.

1 $-3 \times 5 + 6 \div (-2)$

4 $20 \div (-5 - 5)$

2 $-2 \times 7 + 15 \div (-3)$

5 $-3 \times (2 - 7)$

3 $(-4 + 1) \times 5$

6 $(-6 - 2) \div (-4)$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $-3 \times 5 + 6 \div (-2)$ **Example 4** $-12 + (-3) \times (-7) - 2$ **YOU DO** On your own: Set C. Evaluate each expression.

1 $-12 + (-3) \times (-7) - 2$

4 $-20 \div (-2 \times 5)$

2 $-6 + (-6) \div (-2) \times 2$

5 $(-3)^2 - 10$

3 $5 - 3 \times (-2) + 1$

6 $4 \times (-3) - (-2) \times 5$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.

Example 5
$$\frac{7 + (-12) + 6}{2 - 5}$$

Example 6
$$-6 + (-6) \div (-2) \times 2$$

YOU DO On your own: Set D. Fraction bars: work out the top and bottom first.

1
$$\frac{9 - 17}{-4 + 6}$$

3
$$\frac{-3 \times 8}{-6}$$

2
$$\frac{7 + (-12) + 6}{2 - 5}$$

4
$$\frac{-10 - 14}{2 \times (-3)}$$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$$-5 + 2 \times 3$$

Level 2

$$-3 \times (4 - 9)$$

Level 3

$$\frac{-4 \times 6}{3 - 5}$$

Extension · Stretch yourself

Worked example

Insert brackets to make this true: $-2 + 6 \times -3 = -12$

1. Try $(-2 + 6) \times -3$
2. $= 4 \times (-3) = -12 \checkmark$

Answer: $(-2 + 6) \times -3$

YOU DO

On your own: read the worked example, then try these.

- E1 Insert brackets: $8 - 12 \div -4 + 2 = 2$

- E2 Write an expression using -3 , 4 and 5 that equals -17 .

Lesson summary

Missed the lesson? Start here.

Order of operations

1. Brackets first.
2. $\times$ and $\div$ from left to right.
3. $+$ and $-$ from left to right.
4. Fraction bar: work out top and bottom first.

Level 1

$$-5 + 4 \times (-2)$$

1. $4 \times (-2) = -8$
2. $-5 + (-8)$

Answer: -13

Level 2

$$(-3 - 5) \times 2$$

1. $-3 - 5 = -8$
2. -8×2

Answer: -16

Level 3

$$\frac{-18}{2 - 5}$$

1. $2 - 5 = -3$
2. $-18 \div (-3)$

Answer: 6

Extension

$$(-2 + 6) \times -3$$

1. 4×-3

Answer: -12

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LESSON
1.09

Integers on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with integers

Success criteria

I can:

- ☐ enter a negative number using the $(-)$ key
- ☐ use brackets on a calculator
- ☐ check that an answer is reasonable

Key terms Write the definition as your teacher explains it. $(-)$ key

Estimate

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Use your calculator: $28 - 37$ and $-17 + 11$

Example 2

 $[5 + (-20)] \div (-3)$

YOU DO

On your own: Set A. Use your calculator.

1 $28 - 37$

2 $-17 + 11$

3 $-45 - 38$

4 -16×3

5 $-144 \div (-12)$

6 $-250 + 175$

YOU DO On your own: Set B. Use your calculator.

1 $18 - 15 \times 2$

4 $-7 \times (-8) - 60$

2 $-16 \times 3 + (-4)$

5 $-96 \div 8 + 5$

3 $100 - 120 \div 4 + 17$

6 $(-13)^2$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $18 - 15 \times 2$ on a calculator**Example 4** $100 - 120 \div 4 + 17$ **YOU DO** On your own: Set C. Use brackets on your calculator.

1 $[5 + (-20)] \div (-3)$

4 $(-8 + 3) \times (-6 - 1)$

2 $(-14 - 22) \div (-4)$

5 $(-55 + 19) \div 12$

3 $-3 \times (17 - 29)$

6 $-(-7 - 11) \times 2$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $-16 \times 3 + (-4)$ **Example 6** $\frac{-84 - 16}{-25}$ **YOU DO** On your own: Set D. Estimate first, then check with a calculator.1 Estimate -49×21 , then calculate.3 Jo typed -6^2 and got -36 . Why? How do you get 36?2 Estimate $-612 \div 29.8$ (1 d.p.), then calculate.4 Find the error: $20 - 4 \times 3 = 48$ on a calculator?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$-84 + 39$

Level 2

$-15 \times (-7) - 100$

Level 3

$(-35 - 45) \div (-16)$

Extension · Stretch yourself

Worked example

Evaluate $\frac{-48 + 12}{-3 \times (-4)}$ on a calculator.

1. Put brackets around the top and the bottom
2. $(-48 + 12) \div (-3 \times (-4))$
3. $= -36 \div 12$

Answer: -3

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{-125 - 25}{6 \times (-5)}$

E2 $\frac{(-9)^2 - 1}{-16}$

Lesson summary

Missed the lesson? Start here.

Remember 1. Use the $(-)$ key for negative numbers.
3. Use brackets for fraction bars.

2. Calculators follow the order of operations.
4. Estimate to check your answer.

Level 1

$-23 + 9$

1. $(-) 23 + 9 =$

Answer: -14

Level 2

$18 - 15 \times 2$

1. $\times$ first

Answer: -12

Level 3

$(-36 + 6) \div 5$

1. Brackets on top

Answer: -6

Extension

$(-3)^2$ vs -3^2

1. $(-3)^2 = 9$, $-3^2 = -9$

Answer: 9 and -9

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LESSON
1.10

Integer problems

Date _____

Learning intentions

We are learning to:

- solve real problems using integers

Success criteria

I can:

- ☐ decide which operation a problem needs
- ☐ write an integer calculation for a word problem
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Rise

Balance

Deposit

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 The minimum overnight temperature in Goulburn was -5°C . During the day it rose to 11°C . By how much did it rise?

Example 2 A bank account has \$120. Three withdrawals of \$55 are made. What is the balance?

YOU DO On your own: Set A. Temperature problems.

- 1 It was -3°C at 6 am and 14°C at noon. What was the rise?

- 2 It was 8°C and fell 12°C . What is the new temperature?

- 3 The temperature fell 2°C each hour from 5°C . What is it after 4 hours?

- 4 Find the average of -4°C , 2°C , -1°C and -5°C .

YOU DO On your own: Set B. Heights and depths.

- 1 A diver at -18 m rises 7 m. How deep is she?

- 2 A lift goes from level 3 to level -2 . How many floors did it travel?

- 3 A plane at 900 m descends 350 m. What is its height now?

- 4 A cave is 35 m below sea level and a hill is 120 m above. Find the difference.

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

- Example 3** A lift at level 4 goes down 7 floors. Which level is it on?

- Example 4** Balance $-\$35$, deposit $\$120$. New balance?

YOU DO On your own: Set C. Money problems.

- 1 Balance $\$40$, withdraw $\$65$. New balance?

- 2 Balance $-\$30$, deposit $\$75$. New balance?

- 3 Sam owes $\$12$ to each of 5 friends. Write this as an integer and find the total.

- 4 A shop made $-\$250$ on Monday and $\$400$ on Tuesday. Total?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Harrison saves \$80 a week and pays \$210 a month for a car loan. How much after 6 months (26 weeks)?
Example 6 A diver at -42 m rises 6 m every minute. Where is she after 5 minutes?
YOU DO On your own: Set D. Multi-step problems.**1** Harrison saves \$80 a week. Each month \$210 is taken out for a car loan. How much is in the account after 6 months (26 weeks)?
2 A game scores $+5$ for a win and -3 for a loss. Mia wins 7 and loses 9. What is her score?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1It was -6°C and rose 10°C . What is the temperature now?
Level 2A diver goes from -30 m to -12 m. How far did she rise?
Level 3

A score starts at 0. Gain 4, lose 7, lose 7, gain 3. Final score?

Extension · Stretch yourself

Worked example

A submarine at -240 m rises 15 m per minute. How long until it is at -60 m?

- Distance to rise: $-60 - (-240) = 180$ m
- Time: $180 \div 15 = 12$

Answer: 12 minutes

YOU DO

On your own: read the worked example, then try these.

- E1** A freezer is at 4°C and cools by 3°C every 10 minutes. How long until it is -17°C ?

- E2** Write your own integer problem with an answer of -8 .

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read carefully and underline the numbers.
 2. Rise, gain, deposit $\rightarrow$ add. Fall, loss, withdraw $\rightarrow$ subtract.
 3. Difference: larger $-$ smaller.
 4. Check the answer makes sense.

Level 1

From -5°C to 11°C

1. $11 - (-5)$

Answer: 16°C rise

Level 2

\$20 balance, withdraw \$50

1. $20 - 50$

Answer: $-\$30$

Level 3

6 months: $+\$80 \times 26$, $-\$210 \times 6$

1. $2080 - 1260$

Answer: $\$820$

Extension

Rise 15 m/min from -240 m to -60 m

1. $180 \div 15$

Answer: 12 minutes

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